

## Synergistic Activities

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1. **Mentoring emerging climate-data scientists.** Served as an NSF LEAP REU mentor in 2025 and supported undergraduate and master's-level research using neural-network models for climate and environmental hazards. This work helped students turn climate-risk questions into reproducible data workflows, model experiments, uncertainty summaries, and interpretable results.
2. **Teaching deep learning for climate science.** Delivered invited lectures at the University of Florida on deep learning applications in climate science, including CNN, LSTM, and hybrid ConvLSTM approaches for spatiotemporal climate data. These lectures connected machine-learning concepts to climate impacts, geospatial analysis, and responsible use of artificial intelligence for environmental prediction.
3. **Transferring AI methods across climate, hydrology, and air quality.** Developed and published machine-learning workflows for tropospheric ozone forecasting, urban water-demand forecasting, air-dispersion modeling, and environmental-health risk assessment. This cross-domain work supports broader transfer of atmospheric-science methods into decision-relevant environmental data science.
4. **Professional peer review and disciplinary service.** Reviewed manuscripts for interdisciplinary journals including *Scientific Reports*, *Journal of Hydrology*, and *Springer Nature Applied Sciences*. This service contributes to rigorous scholarship at the intersection of climate science, hydrology, geospatial analysis, environmental modeling, and applied machine learning.
5. **Open and reproducible climate-risk workflows.** Built Python, MATLAB, R, and geospatial workflows for large environmental datasets, including NetCDF/GRIB processing, model training, hindcast evaluation, and diagnostic visualization. The proposed fellowship will extend this commitment through transparent evaluation of probabilistic forecasts and reusable workflows for heat-extreme prediction.